EXPERIMENT - 7

7. Perform basic video processing operations on the captured video. Read captured video in python and display the video, in slow motion and in fast motion.

Aim:

To read a captured video in Python using OpenCV and display the video in normal speed, slow motion, and fast motion.

Procedure:

1. Import the OpenCV library.
2. Open the captured video using cv2.VideoCapture().
3. Read and display each video frame.
4. For slow motion, increase the frame display delay.
5. For fast motion, decrease the frame display delay.
6. Press the Esc key to stop the video.

Program:

```
import cv2
cap = cv2.VideoCapture("video.mp4")
while cap.isOpened():
   ret, frame = cap.read()
   if not ret:
      break
   cv2.imshow("Normal Video", frame)
   if cv2.waitKey(30) & 0xFF == 27:
      break

cap.release()

cap = cv2.VideoCapture("video.mp4")

while     cap.isOpened():
   ret, frame = cap.read()
   if not ret:
      break
   cv2.imshow("Slow Motion", frame)
   if cv2.waitKey(100) & 0xFF == 27:
```

```
        break

cap.release()

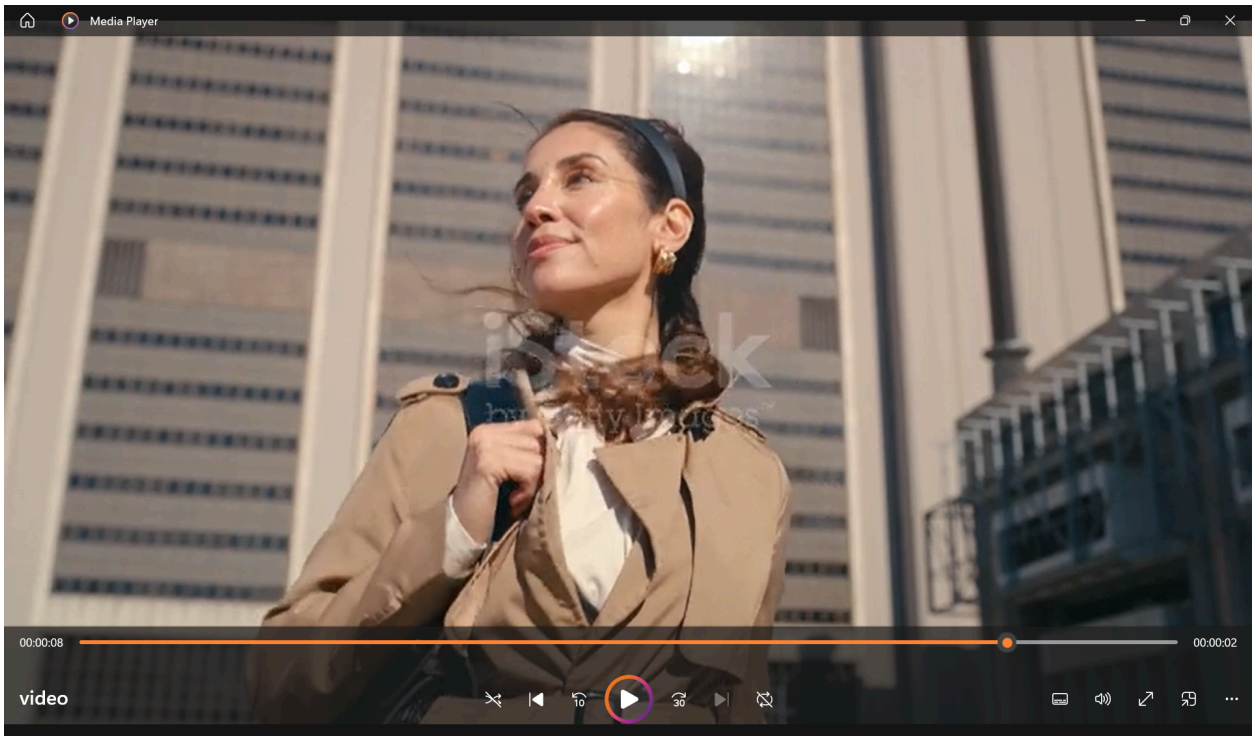
cap = cv2.VideoCapture("video.mp4")

while      cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow("Fast Motion", frame)
    if cv2.waitKey(1) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()
```

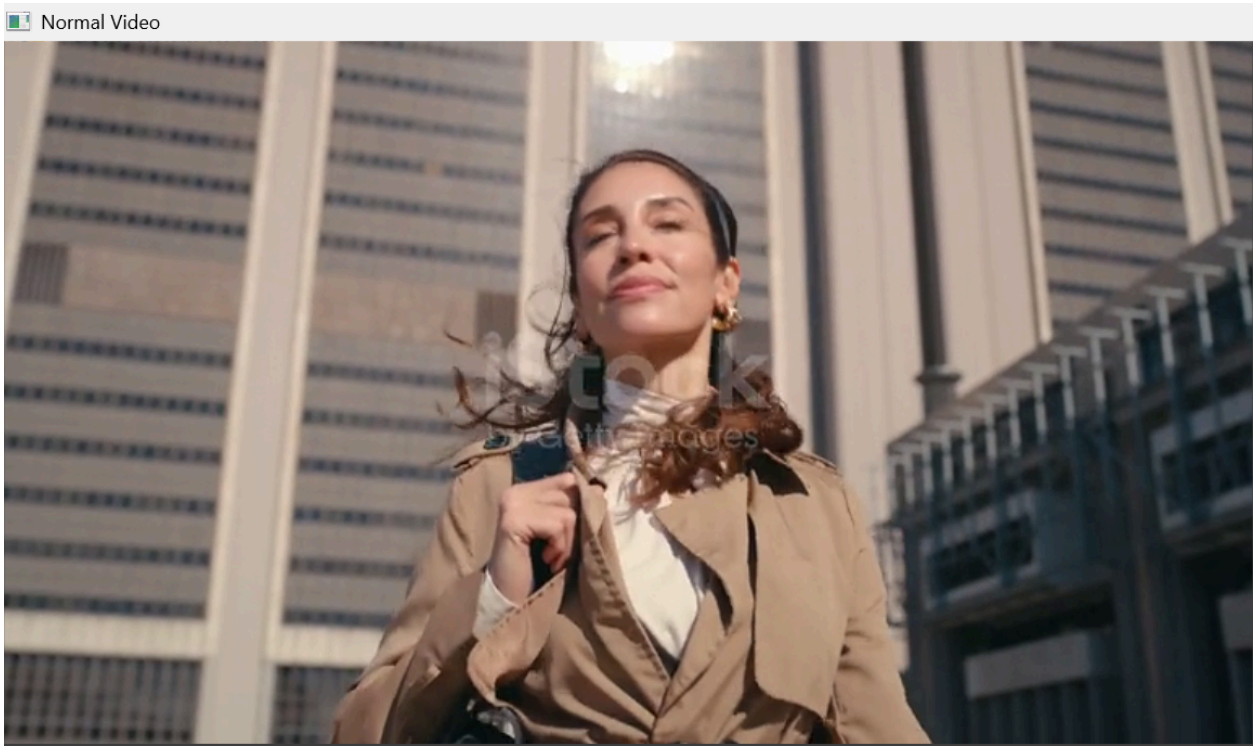
Sample Input:

Video.mp4

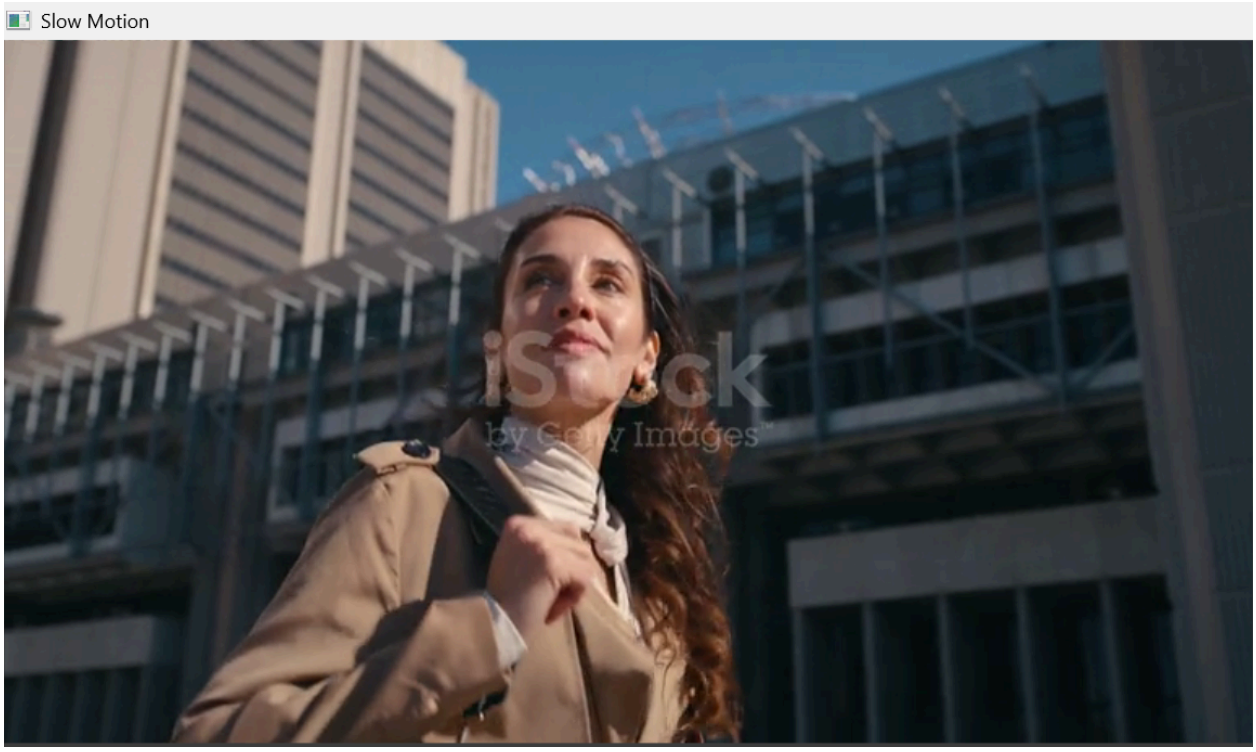


Sample Output:

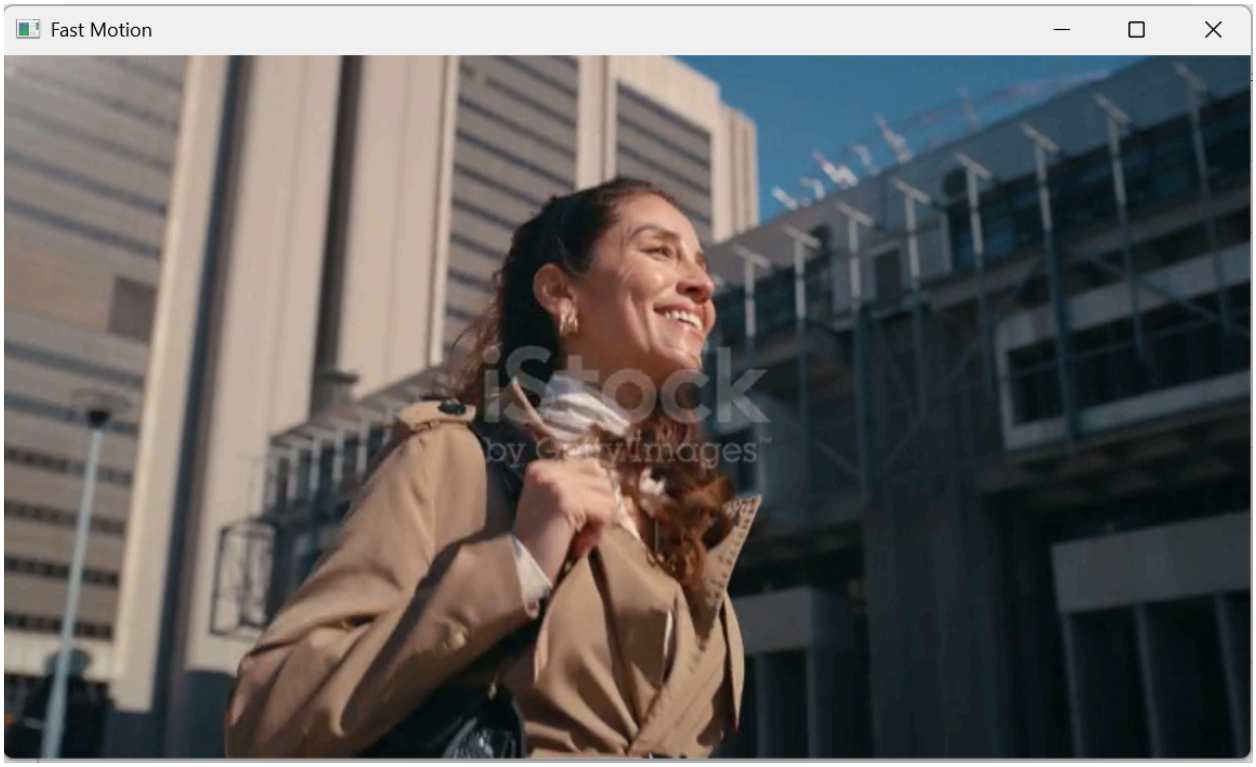
Normal Video: Video plays at approximately normal speed.



Slow Motion: Frames are displayed with a longer delay.



Fast Motion: Frames are displayed with a shorter delay.



Result

Thus, the captured video was successfully read and displayed using OpenCV, demonstrating normal, slow-motion, and fast-motion playback.